
A Note on Mathematical Rigor

Subtleties, Misconceptions, and Pitfalls in Calculus Writing

When writing formal proofs or subjective solutions in analysis and calculus, the presentation of an argument is as critical as the final result. Ambiguous notation, skipped logical steps, and hidden assumptions can invalidate an otherwise correct intuition. This note addresses how to structure functional equation deductions and highlights some of the subtle writing and logical errors that routinely appear in student work.

Motivating Example: Deducing Differentiability

A common hurdle in calculus writing is deciding *when* a function can be declared differentiable directly from a given functional equation. The four cases below, though closely related in appearance, differ profoundly in what they allow.

Case 1: $|f(x) - f(y)| = 2|x - y|$

One cannot directly declare f to be differentiable. The condition forces the difference quotient $\frac{f(x) - f(y)}{x - y}$ to equal either $+2$ or -2 , but without proving the quotient does not “jump” between these values as $x \rightarrow y$, the limit cannot be taken. The valid approach is to deduce the algebraic structure first. Define $g(x) = f(x) - f(0)$.

Step 1: Obtain $g(x)^2$.

Substitute $y = 0$ into the original equation:

$$|f(x) - f(0)| = 2|x| \implies |g(x)| = 2|x|.$$

Squaring both sides gives $g(x)^2 = 4x^2$.

Step 2: Obtain $g(x)g(y)$.

Square the original equation:

$$(f(x) - f(y))^2 = 4(x - y)^2 \quad (*)$$

$$f(x)^2 - 2f(x)f(y) + f(y)^2 = 4x^2 - 8xy + 4y^2.$$

Write $f(x) = g(x) + c$ where $c = f(0)$. Then $f(x)^2 = g(x)^2 + 2cg(x) + c^2$. Substituting and using (*):

$$\begin{aligned} [g(x)^2 + 2cg(x) + c^2] - 2[g(x)g(y) + cg(x) + cg(y) + c^2] \\ + [g(y)^2 + 2cg(y) + c^2] = 4x^2 - 8xy + 4y^2. \end{aligned}$$

After expanding and simplifying:

$$g(x)^2 - 2g(x)g(y) + g(y)^2 = 4x^2 - 8xy + 4y^2.$$

Using $g(x)^2 = 4x^2$ and $g(y)^2 = 4y^2$ yields

$$\begin{aligned} 4x^2 - 2g(x)g(y) + 4y^2 &= 4x^2 - 8xy + 4y^2 \\ \implies -2g(x)g(y) &= -8xy \implies g(x)g(y) = 4xy. \end{aligned} \quad (**)$$

Step 3: Pin down g .

Set $y = 1$ in (**):

$$g(x)g(1) = 4x.$$

From (**) with $x = 1$: $g(1)^2 = 4$, so $g(1) = \pm 2$.

If $g(1) = 2$: then $2g(x) = 4x$, giving $g(x) = 2x$.

If $g(1) = -2$: then $-2g(x) = 4x$, giving $g(x) = -2x$.

Step 4: Recover f .

Since $f(x) = g(x) + f(0)$, we conclude $f(x) = \pm 2x + c$, where $c = f(0)$ is an arbitrary constant. This is trivially differentiable everywhere, with $f'(x) = \pm 2$.

Case 2: $f(x) - f(y) = 2(x - y)$

Here the difference quotient is restricted to exactly one value. Differentiability follows directly from first principles:

$$f'(y) = \lim_{x \rightarrow y} \frac{f(x) - f(y)}{x - y} = \lim_{x \rightarrow y} \frac{2(x - y)}{x - y} = \lim_{x \rightarrow y} 2 = 2.$$

Since the limit exists for all $y \in \mathbb{R}$, f is differentiable everywhere.

Principle. Establish analytical properties before operating on them, unless the limit definition allows immediate evaluation.

Case 3: $|f(x) - f(y)| \leq 2|x - y|$

This says that f is *Lipschitz continuous* with Lipschitz constant $L = 2$. One can immediately deduce continuity (in fact, uniform continuity), but *not* differentiability everywhere.

Why continuity follows.

Fix $\epsilon > 0$. Choose $\delta = \epsilon/2$. Then whenever $|x - y| < \delta$,

$$|f(x) - f(y)| \leq 2|x - y| < 2\delta = \epsilon.$$

This is exactly the ϵ - δ definition of uniform continuity on all of $\mathbb{R}$.

Why differentiability need not hold.

Consider $f(x) = 2|x|$. Then $|f(x) - f(y)| = 2||x| - |y|| \leq 2|x - y|$ by the reverse triangle inequality, so the Lipschitz condition is satisfied. However, f is not differentiable at $x = 0$, since

$$\lim_{h \rightarrow 0^+} \frac{f(h) - f(0)}{h} = 2, \quad \lim_{h \rightarrow 0^-} \frac{f(h) - f(0)}{h} = -2.$$

The left and right derivatives disagree, so $f'(0)$ does not exist.

What can be said about differentiability.

By Rademacher's theorem, every Lipschitz function on $\mathbb{R}$ is differentiable *almost everywhere* — that is, the set of points where f fails to be differentiable has Lebesgue measure zero (for now, you may think of a special type of such measure-zero set: any countable set). Moreover, wherever $f'(x)$ exists, the Lipschitz condition forces

$$|f'(x)| = \lim_{h \rightarrow 0} \left| \frac{f(x+h) - f(x)}{h} \right| \leq 2.$$

So the derivative, when it exists, is bounded by the Lipschitz constant.

Key takeaway. The inequality “ $\leq$ ” in place of “ $=$ ” admits a much wider class of functions. One gains strong continuity and almost-everywhere differentiability, but the algebraic trick of Case 1 (squaring to pin down the exact form) no longer applies. The condition constrains the *size* of the difference quotient but not its *value*.

Case 4: $f(x) - f(y) \leq 2(x - y)$

This is a one-sided inequality without absolute values, and its character is fundamentally different from the previous cases.

A hidden symmetry.

Swapping the roles of x and y in the condition $f(x) - f(y) \leq 2(x - y)$ gives

$$f(y) - f(x) \leq 2(y - x) = -2(x - y).$$

Rewriting: $-(f(x) - f(y)) \leq -2(x - y)$, i.e.

$$f(x) - f(y) \geq 2(x - y).$$

Combining both inequalities:

$$f(x) - f(y) \leq 2(x - y) \quad \text{and} \quad f(x) - f(y) \geq 2(x - y) \implies f(x) - f(y) = 2(x - y).$$

This collapses to Case 2 identically. Thus $f(x) = 2x + c$ for some constant c , and f is differentiable everywhere with $f'(x) = 2$.

Key takeaway. The absence of absolute values is critical. The signed inequality $f(x) - f(y) \leq 2(x - y)$ must hold for *all* $x, y \in \mathbb{R}$, including when $x < y$ (which reverses the sign of the right side). This bidirectional constraint forces equality, and the problem reduces to Case 2. Always test what happens when the roles of x and y are swapped in a universal inequality.

Subtle Pitfalls in Analysis Writing

1. The “Premature Differentiability” Jump

Assuming a function is differentiable just because an equation looks like a derivative, or before proving its structure.

THE TRAP (Student Writes)

Given $f(x + y) = f(x) + f(y) + xy$, we find $f'(x)$ by taking the limit:

$$f'(x) = \lim_{y \rightarrow 0} \frac{f(x + y) - f(x)}{y} = \lim_{y \rightarrow 0} \frac{f(y)}{y} + x.$$

Since $\lim_{y \rightarrow 0} \frac{f(y)}{y} = f'(0)$, we get $f'(x) = x + f'(0)$.

THE REALITY

The assumption that $f'(0)$ exists is unproven. The limit $\lim_{y \rightarrow 0} \frac{f(y)}{y}$ must be shown to converge to a real number first. If the problem does not state that f is differentiable at $x = 0$, this entire chain of equalities is invalid.

2. The “Positive Derivative Implies Local Monotonicity” Fallacy

Assuming that if $f'(c) > 0$, then f must be strictly increasing in some open neighborhood $(c - \epsilon, c + \epsilon)$.

THE TRAP (Student Writes)

Let $f(x) = x + 2x^2 \sin(\frac{1}{x})$ for $x \neq 0$, and $f(0) = 0$. By definition, $f'(0) = 1$. Since $f'(0) = 1 > 0$, the function f is strictly increasing on some interval $(-\delta, \delta)$ around the origin.

THE REALITY

For $x \neq 0$, $f'(x) = 1 + 4x \sin(\frac{1}{x}) - 2 \cos(\frac{1}{x})$. As $x \rightarrow 0$, the cosine term causes $f'(x)$ to oscillate between roughly -1 and 3 . Because $f'(x)$ takes negative values arbitrarily close to 0 , the function f is *never* strictly increasing on any complete neighborhood of the origin.

3. The Continuous Derivative (C^1) Assumption

Assuming that because a function is everywhere differentiable, its derivative f' must be continuous.

THE TRAP (Student Writes)

Let $f(x) = x^2 \sin\left(\frac{1}{x^2}\right)$ for $x \neq 0$, and $f(0) = 0$. The derivative is

$$f'(x) = 2x \sin\left(\frac{1}{x^2}\right) - \frac{2}{x} \cos\left(\frac{1}{x^2}\right).$$

By the Extreme Value Theorem, since f' is the derivative of a continuous function on $[-1, 1]$, $f'(x)$ must be bounded on this closed interval.

THE REALITY

Derivatives do not inherit continuity. As $x \rightarrow 0$, the term $\frac{2}{x} \cos\left(\frac{1}{x^2}\right)$ grows without bound. The derivative $f'(x)$ is unbounded on $[-1, 1]$. An everywhere-differentiable function does not guarantee a well-behaved derivative.

4. The Cauchy Functional Equation Overreach

Solving $f(x + y) = f(x) + f(y)$ and immediately concluding $f(x) = cx$ over $\mathbb{R}$.

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Given $f(x + y) = f(x) + f(y)$ for all $x, y \in \mathbb{R}$. By induction, $f(nx) = nf(x)$ for every natural number n . Setting $x = \frac{p}{q}$, we obtain $f(x) = xf(1)$ for all rational numbers. Therefore, substituting $x = \sqrt{2}$, we get $f(\sqrt{2}) = \sqrt{2}f(1)$.

THE REALITY

One cannot analytically jump from $\mathbb{Q}$ to $\mathbb{R}$. Without explicitly imposing a regularity condition (e.g. f is continuous, monotonic, or locally bounded), the limit of rational sequences approaching $\sqrt{2}$ cannot be evaluated. Wildly discontinuous solutions exist over $\mathbb{R}$ via the Axiom of Choice.

5. The C^∞ Taylor Series Trap

Assuming that if a function is infinitely differentiable everywhere (C^∞), it must equal its Taylor series.

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Let $f(x) = \cos(x) + e^{-1/x^2}$ for $x \neq 0$, and $f(0) = 1$. The Maclaurin series coefficients are $f(0) = 1$, $f'(0) = 0$, $f''(0) = -1$, $\dots$. The series is $1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \dots = \cos(x)$. Since f is C^∞ , it equals its Taylor series. Thus $f(1) = \cos(1)$.

THE REALITY

The actual value is $f(1) = \cos(1) + \frac{1}{e}$. The e^{-1/x^2} term is flat at $x = 0$ (all derivatives vanish there), making it invisible to the Taylor series. Infinite differentiability does not guarantee convergence of the Taylor series to the original function; one must prove the remainder

$$R_n(x) \rightarrow 0.$$

6. The Limit of a Composition Fallacy

Freely pulling the limit operator inside a function, assuming $\lim_{x \rightarrow a} f(g(x)) = f\left(\lim_{x \rightarrow a} g(x)\right)$.

THE TRAP (Student Writes)

Let $g(x) = x \sin\left(\frac{1}{x}\right)$ and let $f(y) = 0$ if $y \neq 0$, with $f(0) = 1$. Evaluate $\lim_{x \rightarrow 0} f(g(x))$.

First, the inner limit: $\lim_{x \rightarrow 0} g(x) = 0$. Therefore, $\lim_{x \rightarrow 0} f(g(x)) = f(0) = 1$.

THE REALITY

As $x \rightarrow 0$, $g(x)$ hits 0 infinitely many times but is nonzero elsewhere. Thus $f(g(x))$ jumps between 0 and 1 as you approach the origin; the limit does not exist. One cannot pass a limit inside a discontinuous function unless $g(x)$ completely avoids the point of discontinuity in a punctured neighborhood.

7. The L'Hôpital's Rule Converse Error

Assuming that if $\lim \frac{f'(x)}{g'(x)}$ does not exist, then $\lim \frac{f(x)}{g(x)}$ must also not exist.

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Evaluate $\lim_{x \rightarrow \infty} \frac{x + \sin(x)}{x}$. This is an $\frac{\infty}{\infty}$ form. Taking derivatives yields $\lim_{x \rightarrow \infty} \frac{1 + \cos(x)}{1}$. Since $1 + \cos(x)$ oscillates, the limit of the derivatives does not exist. By L'Hôpital's Rule, the original limit does not exist.

THE REALITY

L'Hôpital's Rule works only when the limit of the ratio of derivatives exists (finitely or as $\pm\infty$). If it oscillates, the rule is inconclusive. The original limit is easily evaluated algebraically:

$$\lim_{x \rightarrow \infty} \left(1 + \frac{\sin x}{x}\right) = 1 + 0 = 1.$$

8. Confusing Riemann Integrability with Antiderivatives

Assuming that every Riemann-integrable function has an antiderivative.

THE TRAP (Student Writes)

Let $f(x) = 1$ for $x \geq 0$ and $f(x) = -1$ for $x < 0$. Define $F(x) = \int_{-1}^x f(t) dt$. By the First

Fundamental Theorem of Calculus, the derivative of the integral is the original function. Therefore $F'(x) = f(x)$ for all $x \in [-1, 1]$.

THE REALITY

Computing $F(x)$ explicitly yields $F(x) = x - 1$ for $x \geq 0$ and $F(x) = -x - 1$ for $x < 0$. At $x = 0$, the left derivative is -1 and the right derivative is 1 , so $F'(0)$ does not exist. The First FTC explicitly requires f to be *continuous* at x to guarantee that $F'(x)$ exists.

9. Pointwise Convergence vs. Commuting Limits

Assuming pointwise convergence is sufficient to swap a limit with an integral.

THE TRAP (Student Writes)

Evaluate $\lim_{n \rightarrow \infty} \int_0^1 2nx(1 - x^2)^n dx$. The pointwise limit of the integrand as $n \rightarrow \infty$ is 0 for all $x \in [0, 1]$. Pulling the limit inside:

$$\int_0^1 0 dx = 0.$$

THE REALITY

Integrating first via substitution yields $\int_0^1 2nx(1 - x^2)^n dx = \frac{n}{n+1}$. Taking the limit gives $\lim_{n \rightarrow \infty} \frac{n}{n+1} = 1$. The true answer is 1, not 0. The interchange was illegal because the sequence of functions does not converge uniformly on $[0, 1]$.

10. Treating Conditionally Convergent Series as Algebraic Sums

Rearranging terms of an infinite sum arbitrarily without establishing absolute convergence.

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Let $S = 1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \cdots$. Rearranging by grouping one positive term with two negative terms:

$$S = \left(1 - \frac{1}{2} - \frac{1}{4}\right) + \left(\frac{1}{3} - \frac{1}{6} - \frac{1}{8}\right) + \cdots = \left(\frac{1}{2} - \frac{1}{4}\right) + \left(\frac{1}{6} - \frac{1}{8}\right) + \cdots = \frac{1}{2}S.$$

Since $S = \frac{1}{2}S$, we must have $S = 0$.

THE REALITY

The actual sum is $\ln 2$. Because the series of absolute values diverges (the series is only conditionally convergent), Riemann's Rearrangement Theorem applies: rearranging the terms can change the sum to any desired real value, or make it diverge. Infinite series cannot be treated like finite arithmetic unless absolute convergence is established first.

11. The “Asymptotic Flatness” Trap (Vanishing Derivative at Infinity)

Assuming that if a differentiable function approaches a horizontal asymptote as $x \rightarrow \infty$, its derivative must approach zero.

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Evaluate $\lim_{x \rightarrow \infty} f'(x)$ given that $\lim_{x \rightarrow \infty} f(x) = 0$. Since the function's values are stabilizing and becoming arbitrarily close to 0, the graph is flattening out. The slope of a horizontal line is zero, therefore $\lim_{x \rightarrow \infty} f'(x) = 0$.

THE REALITY

A function can converge to a limit while its derivative oscillates wildly or even grows unbounded. Consider $f(x) = \frac{\sin(x^3)}{x}$ for $x > 0$. By the Squeeze Theorem, $\lim_{x \rightarrow \infty} f(x) = 0$. However, applying the quotient and chain rules yields

$$f'(x) = 3x \cos(x^3) - \frac{\sin(x^3)}{x^2}.$$

As $x \rightarrow \infty$, the second term vanishes, but the first term $3x \cos(x^3)$ oscillates with amplitude growing to infinity. The limit of the derivative does not exist, showing that global asymptotic behaviour of a function does not constrain the local behaviour of its derivative.

12. The “Jump Discontinuity in a Derivative” Assumption

Assuming that because piecewise functions can have step discontinuities, derivative functions can also feature jump discontinuities.

THE TRAP (Student Writes)

Let $f(x)$ be defined such that $f'(x) = -1$ for $x < 0$ and $f'(x) = 1$ for $x \geq 0$. This represents a valid derivative function with a jump discontinuity at the origin, corresponding to a sharp corner on the original graph.

THE REALITY

A true derivative function can *never* possess a jump discontinuity. If the left and right limits of a difference quotient disagree (as for $f(x) = |x|$), the derivative simply does not exist at that point; $f'(0)$ is undefined, so the domain of f' has a hole.

Furthermore, by Darboux's Theorem, every derivative satisfies the Intermediate Value Property: if a and b are in the domain of f' , then f' takes on every value between $f'(a)$ and $f'(b)$. A derivative can have essential discontinuities (like $\sin(\frac{1}{x})$ oscillating infinitely near zero), but it can never “step” from one value to another.

13. The “Preservation of Strict Inequalities” under Limits

Assuming that if a strict inequality holds for all terms of a sequence, it must survive in the limit.

THE TRAP (Student Writes)

Given a sequence of continuous functions $f_n(x)$ converging pointwise to $f(x)$ on $[0, 1]$: if $f_n(x) > 0$ for all $x \in [0, 1]$ and all $n \in \mathbb{N}$, then taking the limit as $n \rightarrow \infty$ guarantees $f(x) > 0$.

THE REALITY

Limit operations only preserve *weak* (non-strict) inequalities. Even if a sequence is strictly bounded away from zero at every finite index n , the limit function can collapse to the boundary. A trivial counterexample: $f_n(x) = \frac{1}{n}$. For every $n \geq 1$, $f_n(x) > 0$ strictly, yet the uniform limit is $f(x) = 0$. The strict inequality does not survive the limit process.

14. The “Fading Function” Fallacy for Improper Integrals

Assuming that if $\int_0^\infty f(x) dx$ converges, the integrand must eventually approach zero.

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Let f be continuous and non-negative with $\int_0^\infty f(x) dx = L < \infty$. Because the total area under the curve is finite, the function cannot continue to enclose significant area as x grows large. Therefore $\lim_{x \rightarrow \infty} f(x) = 0$.

THE REALITY

A function can have a finite improper integral while its values grow without bound, provided the “spikes” in its graph become sufficiently narrow. Consider a function built from triangular spikes centred at every integer $n \geq 2$: let the n -th spike have height n and base width $\frac{1}{n^3}$. Each triangle has area

$$\frac{1}{2} \cdot \frac{1}{n^3} \cdot n = \frac{1}{2n^2}.$$

The total area $\sum_{n=2}^\infty \frac{1}{2n^2}$ converges by the p -series test, so the integral is finite. Yet the function reaches height n at every integer, so $\limsup_{x \rightarrow \infty} f(x) = \infty$. The function certainly does not approach zero.

15. The “Everywhere Differentiable Implies Integrable” FTC Trap

Assuming that if $F(x)$ is differentiable everywhere on an interval, one can always evaluate the Riemann integral of its derivative via $\int_a^b F'(x) dx = F(b) - F(a)$.

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Let $F(x) = x^2 \sin\left(\frac{1}{x^2}\right)$ for $x \neq 0$, and $F(0) = 0$. This function is differentiable for all real numbers, so $F'(x)$ exists everywhere on $[0, 1]$. By the Fundamental Theorem of Calculus, since F is the exact antiderivative of F' :

$$\int_0^1 F'(x) dx = F(1) - F(0) = \sin(1) - 0 = \sin(1).$$

THE REALITY

The standard Fundamental Theorem of Calculus for Riemann integrals harbours a crucial, often-forgotten prerequisite: $F'(x)$ must actually be *Riemann integrable* on $[a, b]$. In particular, it must be bounded. Computing the derivative for $x \neq 0$ gives

$$F'(x) = 2x \sin\left(\frac{1}{x^2}\right) - \frac{2}{x} \cos\left(\frac{1}{x^2}\right).$$

As $x \rightarrow 0$, the first term vanishes, but the $\frac{2}{x}$ coefficient in the second term causes oscillations with amplitude growing without bound. Because $F'(x)$ is unbounded near the origin, it is not Riemann integrable on $[0, 1]$. The expression $\int_0^1 F'(x) dx$ is itself undefined in the Riemann sense.